

Practical Laboratory 2

OPNsense Firewall Policy Testing, Logging and Packet Analysis

WADF105 — Network Security Fundamentals

Full name	Mohammed Ibrahim
Student number	C11/26/FCDF/17175
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Analysis Questions

Question 1

Why must the specific block rules be placed above the broad allow rule?

OPNsense reads firewall rules from top to bottom and stops at the first one that matches the traffic. There is already a default rule at the bottom that allows everything through. If you put your block rule below that, the allow rule will match first and the traffic will go through before the block rule ever gets a chance to apply. So the block rule has to sit above the allow rule to be seen first and actually do its job.

Question 2

Which five packet attributes are most useful when explaining a firewall decision?

The five most useful ones are the source IP address, the destination IP address, the protocol being used such as ICMP or TCP, the destination port number where relevant, and the interface the traffic arrived on. Together these tell you exactly where the traffic came from, where it was going, what kind of traffic it was, and which rule it matched. Without all five you cannot fully explain why a packet was allowed or blocked.

Question 3

Why did blocking ICMP not block HTTPS?

Because ICMP and HTTPS are completely different protocols. The block rule was written specifically for ICMP traffic only, so it only catches ping packets. HTTPS runs over TCP on port 443 and is a different kind of traffic entirely. The firewall only matches rules to traffic that fits all the conditions in that rule. Since the HTTPS traffic did not match the ICMP rule it fell through to the default allow rule and went through normally. That is why a targeted block rule only stops the specific traffic it describes and nothing else.

Question 4

What difference did you observe between blocked TCP port 80 traffic and permitted TCP port 443 traffic?

With the blocked HTTP traffic on port 80, the connection never completed. The curl request timed out and in Wireshark you would see TCP SYN packets being sent out repeatedly with no response coming back because the firewall silently dropped them. With HTTPS on port 443, the connection completed properly. In Wireshark you would see the full three-way TCP handshake succeed followed by TLS traffic. The difference between a hanging connection with retransmissions and a clean completed handshake makes it very clear which traffic the firewall was blocking.

Question 5

What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

The Ubuntu client has a private address of 10.10.10.x which is not routable on the public internet. When traffic from that address leaves through the firewall's WAN interface, outbound NAT translates the source address to the firewall's public WAN address. The internet server receiving the traffic sees the WAN address and responds to it. The firewall then translates the reply back and forwards it to the Ubuntu client. Without this translation the internet would not know where to send replies back to because 10.10.10.x addresses are private and not accessible from outside.

Question 6

Why is restoring the original state an important part of a controlled security laboratory?

Because the next activity or the next student using the environment needs to start from a known working setup. If you leave test rules in place and forget about them, future labs might produce confusing results where things seem broken for no obvious reason. In a real work setting the same applies, temporary rules created for testing should be cleaned up so they do not accidentally block legitimate traffic or leave unexpected gaps in security. Restoring the environment at the end is a professional habit that prevents problems down the line.